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ACADEMIC NETWORK FOR A GREEN & INNOVATIVE EUROPE

Co-funded by the
Erasmus+ Programme
of the European Union

STRATEGY AND ACTION PLAN FOR THE TRANSITION OF A SHIPYARD TO A SUSTAINABLE FABRICATION

Project number 2023-1-RO01-KA220-HED-000158031

WEBSITE: www.angie-project.eu

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Why?

Shipyard

- One of the most traditional branches of heavy manufacturing
- Activities rely heavily on large-scale welding and related processes
- High consumer of materials, energy, and water
- High producer of waste

Carbon footprint

Shipyard's footprint (for 1 tone processed steel)

- Metal deposited by welding: 50 kg wire/t (Process: GMAW 70%, SAW 25%, SMAW 5%)
- Specific consumed energy (kWh/kg wire): GMAW 1.5, SAW 1.0, SMAW 2.5 ⇒ 71.25 kWh/t for welding + 20 kWh/t auxiliary (ventilation, fume extraction) ⇒ 91.25 kWh/t total
- GMAW shielding gas (CO_2 pure): ~0.25 m^3 CO_2/kg wire ⇒ at 35 kg GMAW ⇒ 8.75 m^3 CO_2 directly emitted (1 m^3 CO_2 ≈ 1.98 kg CO_2)
- Heat-treatments (reducing distortions) using flame heating: 30 m^3/t
- Internal logistics (using diesel engines): 2 l/t

Calculation

- CO_2 from electricity: 91.25 kWh/t × 0.242 = 22.08 kg CO_2/t
- CO_2 from shielding gas (directly emitted): 8.75 m^3 × 1.98 = 17.33 kg CO_2/t
- CO_2 from natural gas (flame for heat-treatments): 30 m^3 × 1.993 = 59.79 kg CO_2/t
- CO_2 from diesel engines for logistics: 2 l × 2.682 = 5.36 kg CO_2/t

Total shipyard (operational): ~105 kg CO_2/t processed steel

Emission factors

- Electricity EU: 0.242 kg CO_2/kWh.
- Natural gas: 1.993 kg CO_2/m^3.
- Diesel fuel: 2.682 kg CO_2/l.

What?

LCA

Management tool

Identifying potential impact trade-offs

Approach

Levels

- Technological: conventional welding processes carry a high carbon footprint and limited energy efficiency
- Economic: highly resource-intensive, consuming electricity and auxiliary materials in large amounts
- Strategic and Institutional: developing an organizational culture towards sustainable operation

Expecting to…

Carbon footprint monitoring

The personalized carbon footprint of the shipyard must become a central metric

LCA: Life-Cycle Assessment

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Why Sustainability?

ecological	conservation of natural resources and environmental protection
economic	sustainable economic development and long-term financial viability, without compromising natural or social capital
social	equity, inclusion, social cohesion and the general well-being of the population
strategic	the role of integrating sustainability into long-term thinking and planning
institutional	the ability of public organizations and institutions to function efficiently, legitimately and stably, in support of the public interest and sustainable development goals

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Data

- defining clear functional units (for instance 1 m of weld, 1 m of cut, 1 m² of painted surface)
- applying a gate-to-gate system boundary, where direct operations are measured and upstream emissions are added using standardized emission factors
- a structured measurement campaign captures resource use and emissions for each process. Key indicators include electricity (kWh), process gases (m³), filler and flux (kg), paints and solvents (L), and material scrap (kg). Exposure factors such as fumes, VOCs, and noise
- data are then normalized to functional units, converted into CO_2-equivalents and VOC mass through emission factors, and corrected for overspray, rework, or nesting inefficiencies.

VOC: Volatile organic compounds
EPD: Environmental product declaration

Parameter		Default value	Notes
EF electricity	kg CO_2 / kWh	0,35	Grid-specific
EF diesel	kg CO_2 / litre	2,68	Combustion tailpipe. Include upstream if needed
EF O_2	kg CO_2 / m³	0,2	Oxygen production + transport (correct info from supplier EPD)
EF Acetylene	kg CO_2 / m³	1,6	Acetylene production (correct info from supplier EPD)
EF Argon	kg CO_2 / m³	0,9	Argon production (correct info from supplier EPD)
EF CO_2 gas	kg CO_2 / m³	1,9	CO2 supply chain footprint
EF Compressed air	kg CO_2 / m³	0,05	Compressor electricity upstream
EF Filler wire	kg CO_2 / kg	6	Steel filler wire cradle-to-gate
EF Flux	kg CO_2 / kg	2,5	SAW flux cradle-to-gate
EF Paint	kg CO_2 / litre	3,5	Coatings (solvent-borne) cradle-to-gate
EF Solvent	kg CO_2 / litre	2	Thinners/solvents
EF Coolant	kg CO_2 / litre	1,5	Emulsion/coolant production
EF Steel plate	kg CO_2 / kg	2	Correct mill EPD (naval grades) to be used in analysis
EF Al plate	kg CO_2 / kg	8	If using recycled Al, replace with lower EF
VOC factor	kg VOC / litre paint	0,42	Adjust to product SDS/EPD
Noise factor weight	-	0,5	Relative penalty for noise exposure
Fume factor weight	-	1	Relative penalty for welding fumes
VOC hazard weight	-	1,2	Relative penalty for solvent/VOC exposure

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Sustainability matrix

- for creating a holistic understanding of the challenges and opportunities associated with the transition
- Importance: its ability to provide decision-makers with a transparent and comparative view across different processes
- serves as a foundation for setting priorities, evaluating alternative technologies, and designing targeted interventions that are both realistic and impactful

VOC: Volatile organic compounds

Process / Issues	Technological	Environmental	Economic	Social
Thermal cutting (oxy-fuel, plasma, laser)	High gas and energy consumption; variable precision; losses from non-optimized cutting	CO_2, NO_x, metal fumes; slag difficult to handle	High costs of technical gases; scrap from defective cuts	Exposure to radiation, fumes, noise
Plastic deformation (pressing, rolling)	Requires large presses; fast equipment wear	Waste of hydraulic oils; high energy consumption	High maintenance costs; losses due to geometric defects	Exposure to vibration and noise; poor ergonomics
Machining (groove preparation, finishing)	Tool wear; critical precision for weld grooves	Metal chips hard to recycle; polluting coolants	High cost of tools and fluids; rework losses	Exposure to aerosols, dust and noise
Welding (MAG, SAW, SMAW, hybrids)	Conventional processes with low efficiency; lack of digital monitoring	High CO_2 footprint; metal fumes, slag; intensive gas and filler consumption	High costs for energy, gases, filler; frequent scrap and rework	Exposure to fumes and UV radiation; occupational disease risk
Heat treatment / stress relief	Flame heating → low control; alternative technologies scarcely used	CO_2 and NO_x from fossil fuel combustion	High fuel costs; production downtime	Exposure to heat and flue gases
Painting / anticorrosion	Material losses by overspray; solvent dependency	VOC emissions; hazardous waste (solvents, filters, residues)	High costs of materials and ventilation systems	Exposure to toxic solvents; respiratory and dermatological risks

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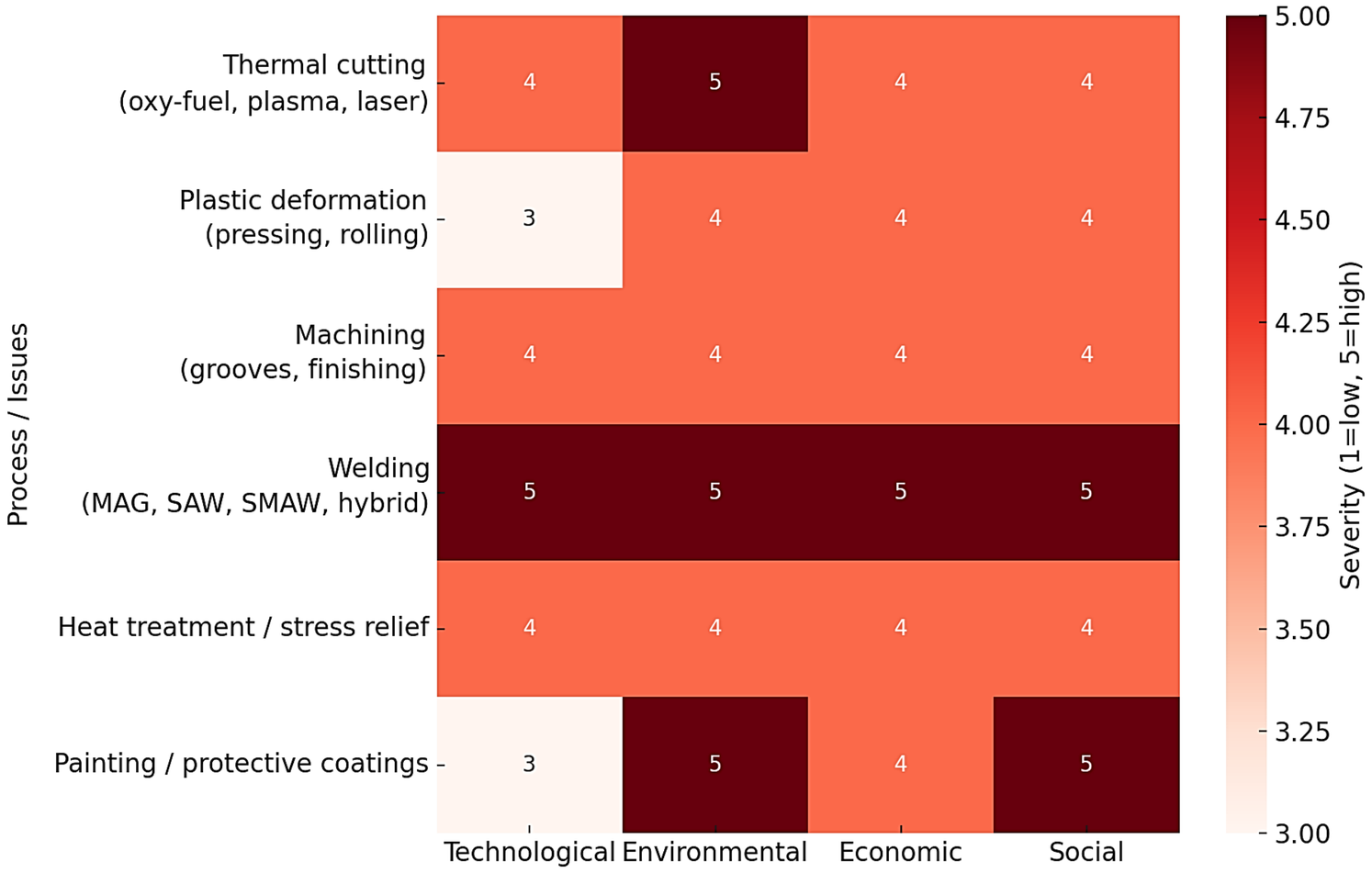

Scoring principle
1 = very low/negligible impact; 2 = low impact; 3 = moderate impact; 4 = high impact; 5 = very high/critical impact

VOC: Volatile organic compounds

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Analysis

Three shipyards

- Subjects: amount of CO_2 emissions (CO_2e/kg per FU and t CO_2e/year), towards material waste (kg/FU, t/year), towards volatile organic compounds and towards energy consumption (kWh/FU, MWh/year)
- Consumption indicator *CO_2e/FU = sum (Consumption x Emission factor)* for electricity, fuels, shielding gases, addition, flux, paint, solvents, cooling, steel/Al scrap
- VOC indicator: *VOC = paint [liters] × VOC factor + solvents [liters]*.
- energy indicator: *Energy [kWh] = electricity + compressed air equivalence*.

Process	Functional unit	Production volume / year	Electricity kWh / FU	Diesel litre / FU	O_2 m^3 / FU	Acetylene m^3 / FU	Argon m^3 / FU	CO_2 m^3 / FU	Compressed air m^3 / FU	Filler wire kg / FU	Flux kg / FU	Paint litre / FU	Solvent litre / FU	Coolant litre / FU	Steel plate scrap kg / FU	Al plate scrap kg / FU	Fume exposure index 0to1	VOC exposure index 0to1	Noise exposure index 0to1
Thermal cutting (oxy-fuel, plasma, laser)	m of cut	500000	0.6	0	0.8	0.08	0	0	0.5	0	0	0	0	0	0.15	0	0.6	0	0.8
Plastic deformation (pressing/rolling)	operation	50000	5	0	0	0	0	0	0.2	0	0	0	0	0	0.02	0	0.1	0	0.6
Machining (groove prep/finishing)	m of groove	300000	0.8	0	0	0	0	0	0.1	0	0	0	0.02	0.05	0.3	0	0.2	0.4	0.7
Welding (MAG/SAW/ SMAW/hybrids)	m of weld	400000	2.5	0	0	0	0.5	0.6	0.1	0.8	0.2	0	0	0	0.1	0	0.9	0.1	0.8
Heat treatment / stress relief	operation	10000	2	4	0	0	0	0	0	0	0	0	0	0	0	0	0.4	0	0.3
Painting / anticorrosion	m^2 coated	200000	0.4	0	0	0	0	0	0.3	0	0	0.25	0.05	0	0	0	0.1	0.9	0.3

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- energy indicator: *Energy [kWh] = electricity + compressed air equivalence*.

Process	VOC, kg / FU	VOC, kg / year
Machining (groove preparation/finishing)	0.02	6000.0
Painting / anticorrosion	0.155	31000.0

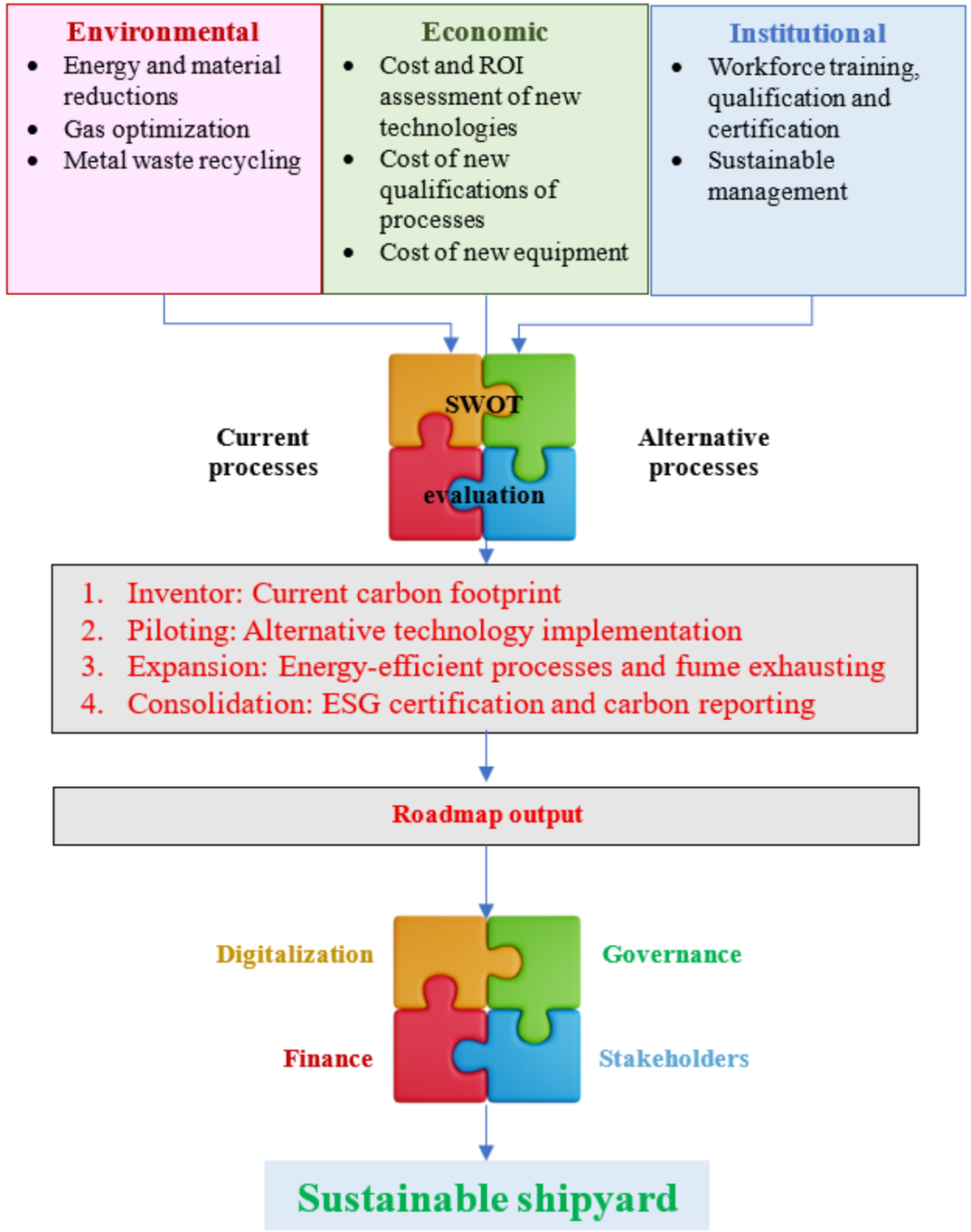
Process	FU-count for 100t of built ship	CO_{2e}, kg	VOC, kg	Energy, kWh	Material waste, kg
Welding	10000.0	79700.0	0.0	25142.86	11000.0
Thermal cutting	12000.0	9876.0	0.0	8057.14	1800.0
Machining	8000.0	8000.0	160.0	6514.29	2400.0
Painting	3500.0	3955.0	542.5	1550.0	0.0
Heat treatment	5.0	57.1	0.0	10.0	0.0
Plastic deformation	20.0	36.0	0.0	100.57	0.4

Records:
the emissions and resources for 100 t of welded construction, including all operations (cutting, forming, machining, welding, heat treatment, painting) are:

- total CO_2e: ~101.6 t
- total VOC: ~0.70 t
- total energy: ~41.4 MWh
- material waste (filler/flux/scraps): ~15.2 t.

Strategy

- from a strategic perspective, this transition requires embedding sustainability metrics directly into shipyard governance.
- the alignment of workforce training, supply chain partnerships, and digitalization initiatives ensures that sustainability is not pursued in isolation but integrated into the competitive fabric of the shipyard's business model.
- the pathway to a sustainable shipyard is a multi-layered transformation (Figure 3): reducing emissions and resource consumption at the source, digitalizing process monitoring, protecting workers' health, and reconfiguring supply chains around lower-carbon inputs.



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Strategy

Aspects to be approached during the piloting stage

Measure	Capex*	Opex**	Impact	KPI	Owner	Timeline
Optimize WPSs and synergic modes	Low	Medium	-10 t CO2e; -5 MWh; -1 t waste	kWh/m; % rework	Welding engineer	0-6 months
Optimize shielding gas and parameters	Low	Low	-4 t CO2e; -0.3 t waste	l/min gas; defects/10m	Welding engineer	0-6 months
Robotized welding cells	High	Medium	-8 t CO2e; -7 MWh	OEE; rework %	Automation team	6-18 months
Laser/Hybrid welding	High	High	-15 t CO2e; -10 MWh	kWh/m; filler use	Tech development	18-36 months
Advanced nesting for cutting	Low	Low	-2.5 t CO2e; -1.2 t waste	Yield %; defects	Cutting manager	0-6 months
Replace oxyfuel with plasma/laser	Medium	High	-5 t CO2e; -2 MWh	kWh/m; kerf losses	Cutting manager	6-18 months
Coolant and tool management	Low	Low	-0.2 t VOC; -0.2 t waste	l/m coolant; tool life	Machining manager	0-6 months
Spray gun optimization (painting)	Low	Low	-0.15 t VOC; -1.5 t CO2e	l/m² paint; overspray %	Paint manager	0-6 months
High-solids / waterborne paints	Medium	Medium	-0.3 t VOC; -1.5 t CO2e	VOC/m²	Paint manager	6-18 months
Compressed air leak management	Low	Low	-0.8 t CO2e; -2 MWh	m³/min leaks; kWh/day	Maintenance	0-6 months
Electrify heating & PPA contract	High	High	-10 t CO2e	EF grid; % electrified	Energy manager	18-36 months
LEV / on-gun extraction	Medium	Medium	-2 t CO2e; rework -15%	mg/m³ fumes; rework %	HSE manager	0-6 months
WAAM for repairs	High	Medium	-6 t CO2e	kg deposition vs. new	R&D team	18-36 months

*** Capital expenditure**
**** Operational expenditure**

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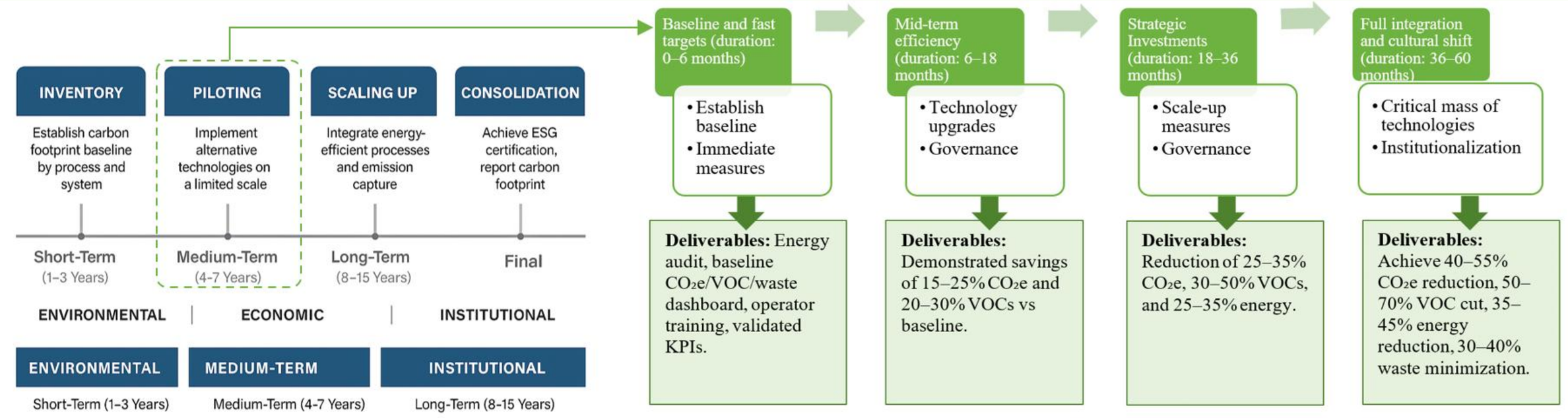
Strategy

Potential targets of the pilot action

Area	Measure	KPI	Baseline	Target
Welding	WPS optimization	kWh/m weld	1,8	1,4
Welding	WPS optimization	Filler kg/m weld	0,45	0,38
Welding	WPS optimization	% rework	12	6
Welding	Gas optimization	Gas flow l/min	18	14
Welding	Gas optimization	Spatter % mass	8	4
Welding	Robotized welding	kWh/m weld	1,8	1,2
Welding	Robotized welding	% rework	12	5
Cutting	Advanced nesting	Yield %	82	88
Cutting	Advanced nesting	% edges reworked	10	5
Cutting	Plasma/laser replace oxyfuel	kWh/m cut	1,5	1
Cutting	Plasma/laser replace oxyfuel	Kerf loss kg/m	0,9	0,5
Machining	Coolant management	Coolant L/m	0,35	0,25
Machining	Coolant management	Tool life (m/tool)	40	60
Painting	Spray housekeeping	Paint L/m²	0,32	0,25
Painting	Spray housekeeping	Overspray %	35	20
Painting	High-solids paints	VOC kg/m²	0,2	0,1
Energy	Leak management	Air leaks m³/min	3,5	1
Energy	Leak management	kWh/day compressors	450	350
Energy	Electrify heating	Grid EF kg CO_2/kWh	0,35	0,2
Health, safety and quality	LEV / on-gun	Fume mg/m³	3	1
Health, safety and quality	LEV / on-gun	% rework	12	8

Strategy


INVENTORY
Establish carbon footprint baseline by process and system
PILOTING
Implement alternative technologies on a limited scale
SCALING UP
Integrate energy-efficient processes and emission capture
CONSOLIDATION
Achieve ESG certification, report carbon footprint
Short-Term (1–3 Years)
Medium-Term (4-7 Years)
Long-Term (8–15 Years)
Final
ENVIRONMENTAL
ECONOMIC
INSTITUTIONAL
ENVIRONMENTAL
MEDIUM-TERM
INSTITUTIONAL
Short-Term (1–3 Years)
Medium-Term (4-7 Years)
Long-Term (8-15 Years)
Baseline and fast targets (duration: 0–6 months)
• Establish baseline
• Immediate measures
Deliverables: Energy audit, baseline CO₂e/VOC/waste dashboard, operator training, validated KPIs.
Mid-term efficiency (duration: 6–18 months)
• Technology upgrades
• Governance
Deliverables: Demonstrated savings of 15–25% CO₂e and 20–30% VOCs vs baseline.
Strategic Investments (duration: 18–36 months)
• Scale-up measures
• Governance
Deliverables: Reduction of 25–35% CO₂e, 30–50% VOCs, and 25–35% energy.
Full integration and cultural shift (duration: 36–60 months)
• Critical mass of technologies
• Institutionalization
Deliverables: Achieve 40–55% CO₂e reduction, 50–70% VOC cut, 35–45% energy reduction, 30–40% waste minimization.

Conclusions

The current state of shipyard manufacturing, dominated by energy-intensive welding, oxyfuel cutting, machining, solvent-based painting, and fossil-fuel heating

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high CO_2e footprint, significant VOC emissions, inefficient energy use, and substantial material waste. The transition to sustainable shipbuilding is therefore not optional but necessary to remain competitive, compliant, and resilient in a global low-carbon economy.

The strategic measures identified—covering welding optimization, gas efficiency, robotization, hybrid/laser processes, nesting and cutting upgrades, VOC reduction in painting, compressed-air and energy management, and long-term adoption of WAAM and electrification—represent a balanced portfolio.

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30–40 tonnes CO_2e savings per 100 tonnes of welded construction within three years, and more than 50% emission reduction within five years.

The action plan structures these measures into four phases:
(1) establishing the baseline and quick wins,
(2) medium-term efficiency through technology pilots,
(3) strategic investments and digitalization,
(4) full integration and cultural transformation.

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